Temperature-Compensation for Magnetostrictive
Corrosion Detection Sensor

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Abstract

Pipelines in the modern day are vital components in not only the world economy, but also in rising demand in energy. As pipelines stretch through harsh, possibly highly corrosive, environments, it is important to perform preventative maintenance to avoid corrosion and damage to its integrity. As modern technology to detect corrosion in pipeline systems is very costly and extensive in size, a new method that is compact and portable is in demand. In this report a magnetostrictive corrosion sensor will be analyzed and suited with a temperature compensating component. From the results of this experiment, the sensing unit, along with the figures crested, can measure the thickness of a steel coupon from the strain measured due to response from a magnetic field while compensating for strain caused by temperature effects.

Introduction/Project Description

The purpose of the experiment was to account for the temperature change when measuring the magnetic field through a given area. In the experiment two different types of rods were used to accomplish this. The first rod was made of Terfenol-D which was selected due to its magnetostrictive properties. The second rod was made of Inconel Alloy 22 which was chosen due to its similar thermal properties as Terfenol-D. Then a tray was designed to hold both rods securely without restricting expansion due to magnetic flux passing through the rods. The testing device consisted of two magnets placed approximately the width of the tray apart attached to a handle that held the magnets in place. To measure the difference in strain between the rods caused by magnetic flux we used a HYPERION si155 [1]. The base form of the equation used to calculate the strain can be seen in equation (1) which was converted to (1e6*sensordata)/Fg to use data obtained from the sensor, where Fg = 0.833. Next an FBG sensor was attached to both rods using adhesive. The change in wavelength due to the elongation of the fiber is what allowed for the measurement of strain on either rod. The initial test was done under no magnetic strain to verify that a change in strain correlated with a change in temperature. Then tests for six different coupons at varying thickness were performed to observe the increase in strain due to the flux passing through the rod. All measurements were tested for at least two days in the same location so the primary variable in the experiment was temperature change in the room. One consideration was to make sure that fibers did not move during testing as this could cause a false increase in strain. The overall purpose of this experiment was to determine the amount of strain that may be detected or caused by changes in temperature.

$$Difference\ in\ strain\ = (1 \times 10^{6})(\frac{(\lambda-\lambda_0)}{\lambda_0}) \tag{1}$$

Measurement Theory

The two metals used in this experiment are Terfenol-D and Inconel Alloy 22. Terfenol-D was chosen to be used in this experiment as it has the largest room temperature magnetostriction of any known material. In other words this metal is known to change shape, introducing strain, in response to a magnetic field. Inconel Alloy 22 was the second metal chosen to be used in this experiment as it is a fully austenitic alloy, meaning that it is non-magnetic. It also has thermal properties that are similar to that of Terfenol-D. It was important to choose a metal that is non magnetic to be able to measure the strain that is solely caused by a change in temperature. To measure the strain caused solely by the introduction to a magnetic field the difference between the strain in the Terfenol-D and Inconel Alloy 22 will be taken. With the Inconel Alloy 22 rod, the strain due to thermal effects can be measured and compensated for in the measured strain of the Terfenol-D rod.

The thermal response of the Inconel Alloy 22 rod needed to match that of the Terfenol-D rod, or in other words, the time it takes for the rods to change temperatures. The thermal response of the two metals was matched by altering the size of the Inconel Alloy 22 rod and keeping that of the Terfenol-D constant. In order to match the transient thermal response of the Inconel Alloy 22 and Terfenol-D rod, calculations of transient heat conduction in lumped systems was conducted.

As a control, the Terfenol-D rod was set to a width of 4 mm with a fixed length of 35 mm. The thermal response of the two metals will be matched by altering the width of the Inconel Alloy 22 rod. The Rayleigh number was calculated for a cylindrical rod of the same length as the Terfenol-D rod with equivalent surface areas using an equivalent diameter for width as seen in equation (2). Taking inputs of the Rayleigh number and Prandtl number, a property of air at a given temperature, a Nusselt number correlation for natural convection over a horizontal cylinder was found to calculate the average heat transfer coefficient as seen in equation (3). With inputs of surface area, metal density, volume, specific heat [2, 3], ambient temperature, and initial temperature, the temperature of the rod as a function of time can be graphed using equation (4) [4]. As graphed in figure 1 the thermal response of the Inconel Alloy 22 matched that of the Terfenol-D when the width of the Inconel Alloy 22 was to 3.85 mm. In the case of this experiment, as the difference between the two rods was minimal, they were both taken to be machined with a width of 4 mm as seen in figure 2. The MATLAB code used to plot the thermal response can be seen in Figure A1.

$$Ra_D = \frac{gB(T_a - T_s)*}{v^2} * Pr \qquad (2)$$

$$Nu = \{0.6 + \frac{0.387 Ra_D^{1/6}}{[1+(\frac{0.559}{Pr})^{9/16}]^{8/27}}\}^2 \text{ for } Ra_D \leq 10^{12} \qquad (3)$$

$$\frac{T(t) - T_\infty}{T_i - T_\infty} = e^{-\frac{hA_s}{\rho V c_p}t} \text{ for } Bi \leq 0.1 \qquad (4)$$

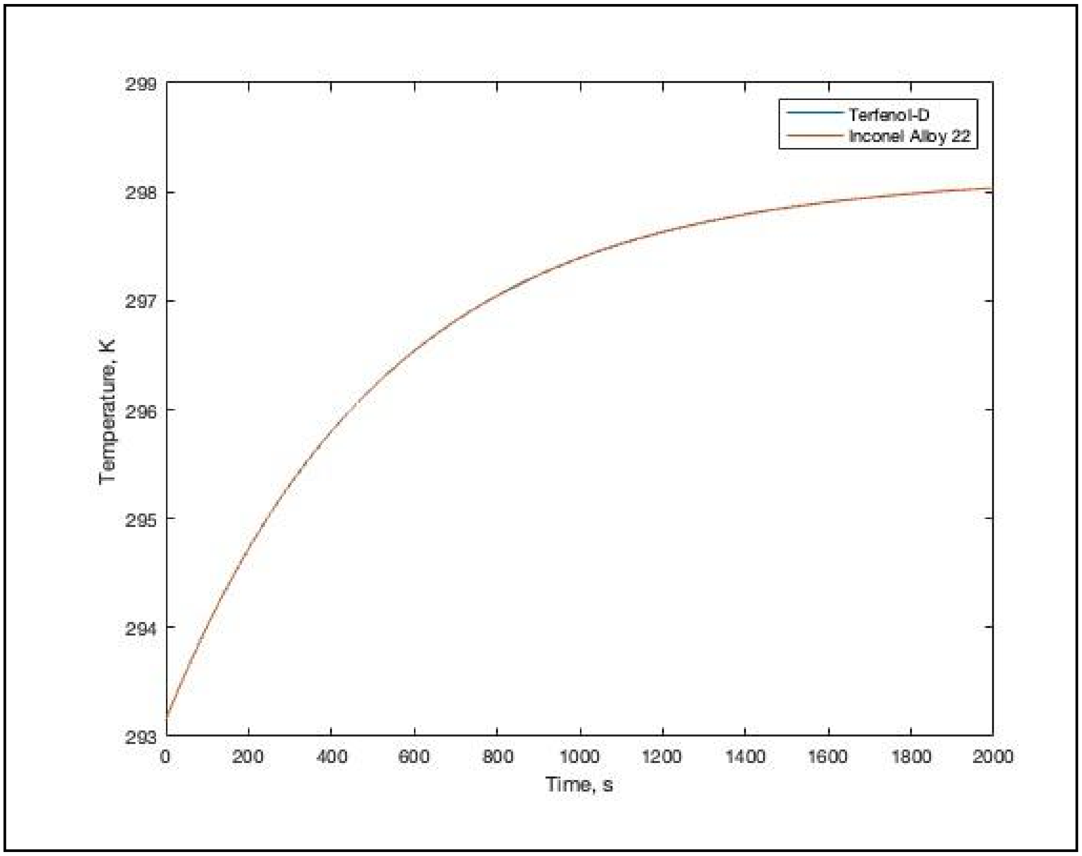

Figure 1: Graph of the matched thermal response of the two metals.

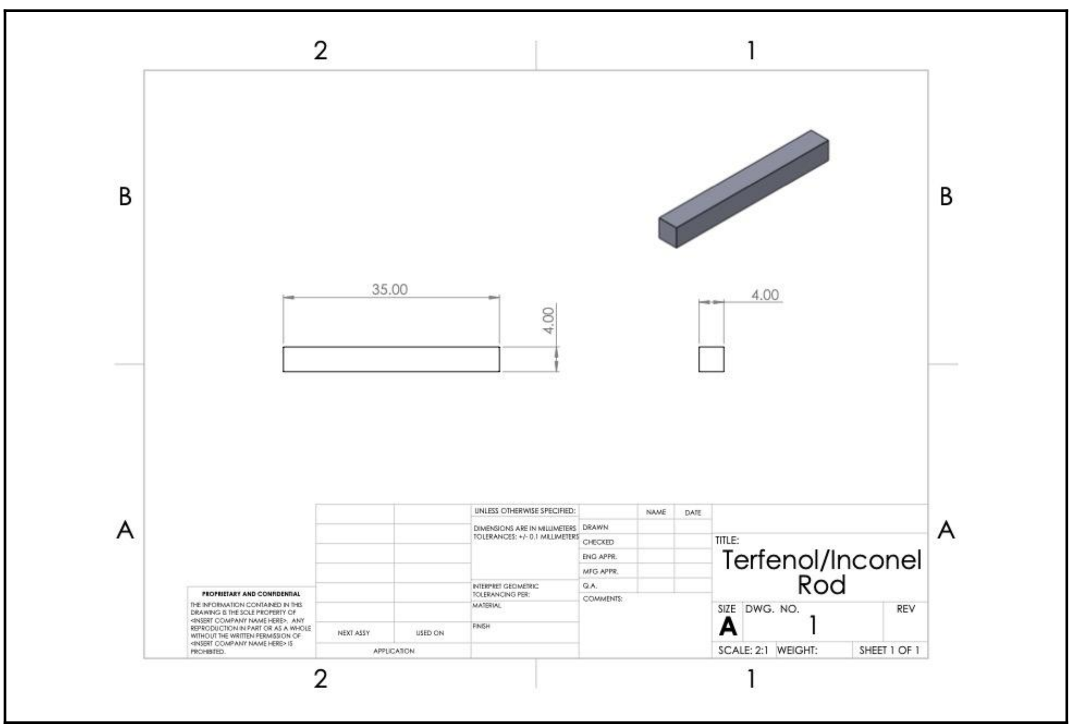

Figure 2: Drawing showing the dimensions of the two metal rods.

To measure the strain experienced by the two metal rods, fiber optic strain gauges, Luna os1100 FBG strain gauges [5], were attached onto the rods with an adhesive to hold them in place. As the strain along the length of the rod was to be measured, the fiber optic cable was carefully aligned and attached along the length. As the rods would need to be moved around, the fiber optic cable needed to be held with care. The adhesive was then given a day to fully solidify. The fiber optic strain measurements were read and analyzed using a Luna si155 HYPERION Optical Sensing Instrument [1]. The sensing instrument was run at a 100 Hz sampling rate.

Experimental Setup

Before putting the metal samples in the tray, strain gauges had to be attached to each one. The strain gauges were attached to the metal samples using super glue as an adhesive. They were laid lengthwise across the entirety of the 35mm samples. After a full day of letting the adhesive dry, the metal samples were secured in the slots of the tray with the clip. A piece of foam was put between the clip and the metal samples so as to not disrupt the fragile strain gauges. Then, before adding the tray to the apparatus, the coupon needed to be attached. On the first test, the thickest coupon was used. Two members of the group with safety equipment such as gloves and safety glasses slowly lowered the magnet onto the coupon. The magnets used were very strong so all metal objects in the area or in pockets had to be removed from the area. After carefully attaching the magnets to the coupon, the data acquisition was started. Then, the tray with the metal samples and strain gauges was slid into the bottom slot of the apparatus, to get it as close to the coupon as possible. This was left for two days to collect data. After the 2 days, the tray was removed and then the data acquisition was stopped. This process was repeated for all six coupons, resulting in almost two weeks of data acquisition. For each experiment, a graph was made from the data to show the difference in strain between Terfenol-d and Inconel Alloy 22.

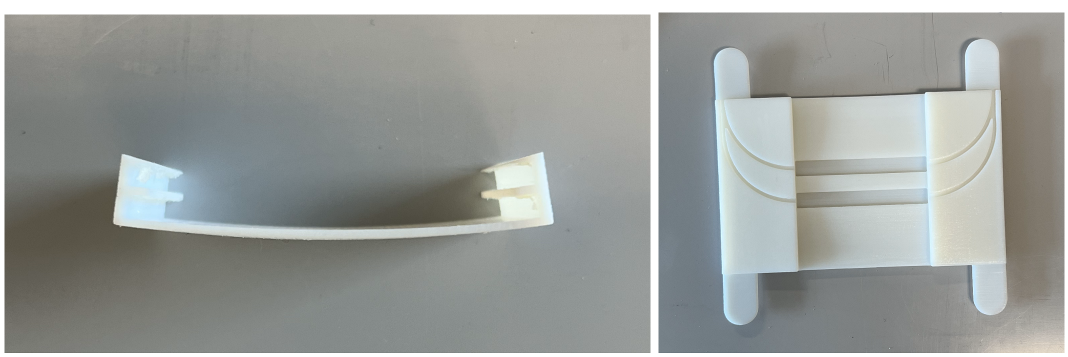

Figure 4: Final Tray and Clip Iterations

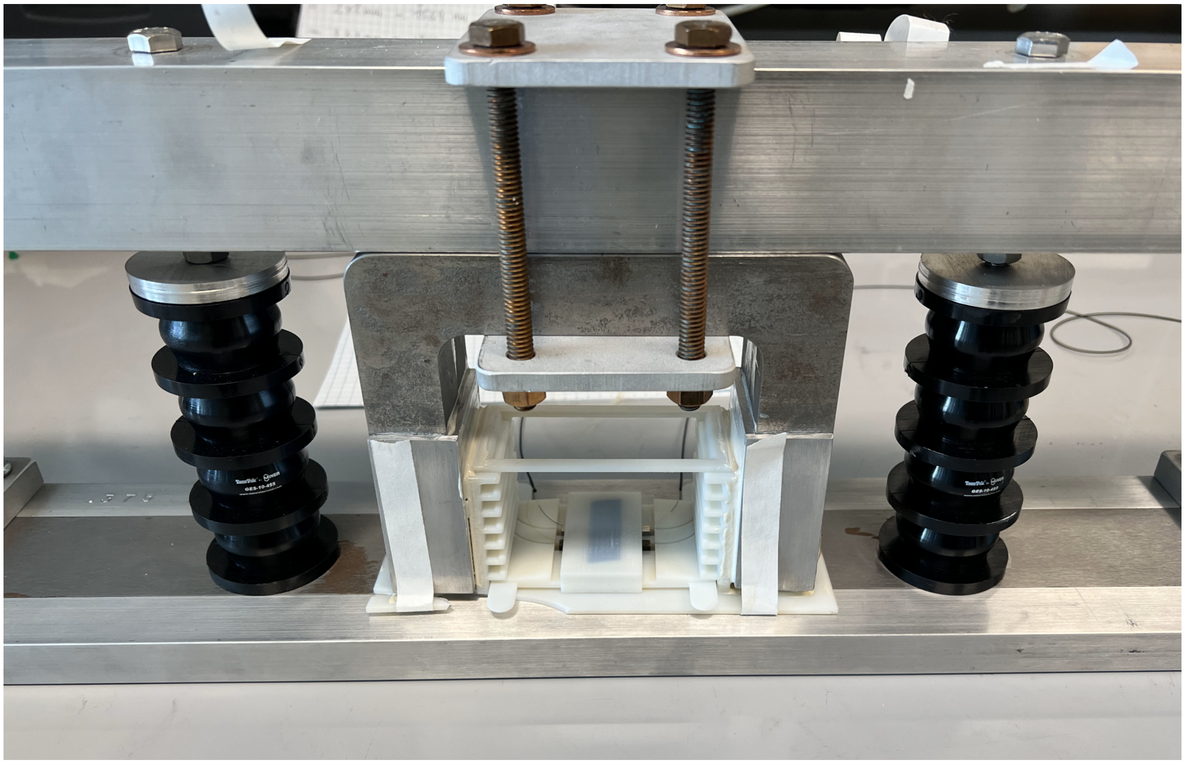

Figure 5: Experimental Apparatus Setup Complete

Uncertainty Analysis

For this experiment the sources identified as the source of potential type B uncertainty were the Hyperion si-155 interrogator and the Luna os1100. For the Hyperion si-155 two sources of uncertainty were identified. wavelength reliability and wavelength accuracy. Both factors were 1 picometer. The FBG being used had a sensitivity of 1.2 picometer per macrostrain. This value was used as a conversion factor in conjunction with the Hyperion si-155 values to calculate overall type B uncertainty. Equation (5) shows the formula used for the calculations.

For type A uncertainty the standard deviation was taken for the difference in strain between the two rods for each coupon. This resulted in a different type A uncertainty for each experiment. The total uncertainty was calculated by standard means of adding the squares for the different types of uncertainty. The equation used in calculations was equation (6) where u_a^2 is different for each coupon resulting in 6 different u_c values. The MATLAB code used to calculate these values can be found in Figure A2.

$$u_b = 1.2 \times \sqrt{u_1^{\ 2} + u_2^{\ 2}} \tag{5}$$

$$u_c = \sqrt{u_a^{\ 2} + u_b^{\ 2}} \tag{6}$$

Results and Discussion

Through this experiment the uncertainty in the measurement of strain was found with measurement compensation due to temperature effects. From using a newly designed tray to hold multiple rods, measurements of the strain due to a magnetic field as well as the strain due to temperature effects were found. The results of the calculations of the total uncertainty are compiled in table 1 for each coupon size. Table 2 shows the 95% confidence measurement made for each coupon. The average difference in strain between the two rods was plotted against the coupon thickness as seen in figure 6. With this graph the sensing unit can be used and with a measured difference between the two rods, the thickness of the coupon can be found. During each measurement a plot of the temperature versus strain for each metal rod was generated for each testing coupon size used. These graphs can be seen in figures A3-A8. The difference in the strain plotted against time can be found in figure A9 and was used to generate the plot of the average strain difference vs thickness. To improve on this experiment, similar adhesives could have been used to avoid any strain measurement uncertainties. Along with similar adhesives, a longer curing time could have been used to ensure the measured strain does not have any effects from the adhesive. From this experiment, the sensing unit along with the calculation and results found can measure the thickness of a steel coupon from the strain measured due to response from a magnetic field while compensating for strain caused by temperature effects.

Table 1: Uncertainty values for each coupon in microstrain

Coupon Size, inch	Type B Uncertainty, microstrain	Type A Uncertainty, microstrain	Total Uncertainty, microstrain
0.500	1.1785	1.2688	1.7317
0.375	1.1785	3.6491	3.8347
0.250	1.1785	4.8944	5.0343
0.187	1.1785	2.7215	2.9657
0.125	1.1785	0.8245	1.6915

Table 2: 95% Confidence for each coupon in microstrain

Coupon Size, inch	95% Confidence, microstrain
0.500	16.12 ± 3.46
0.375	103.2 ± 7.67
0.250	265.9 ± 10.07
0.187	367.4 ± 5.93
0.125	406 ± 2.88

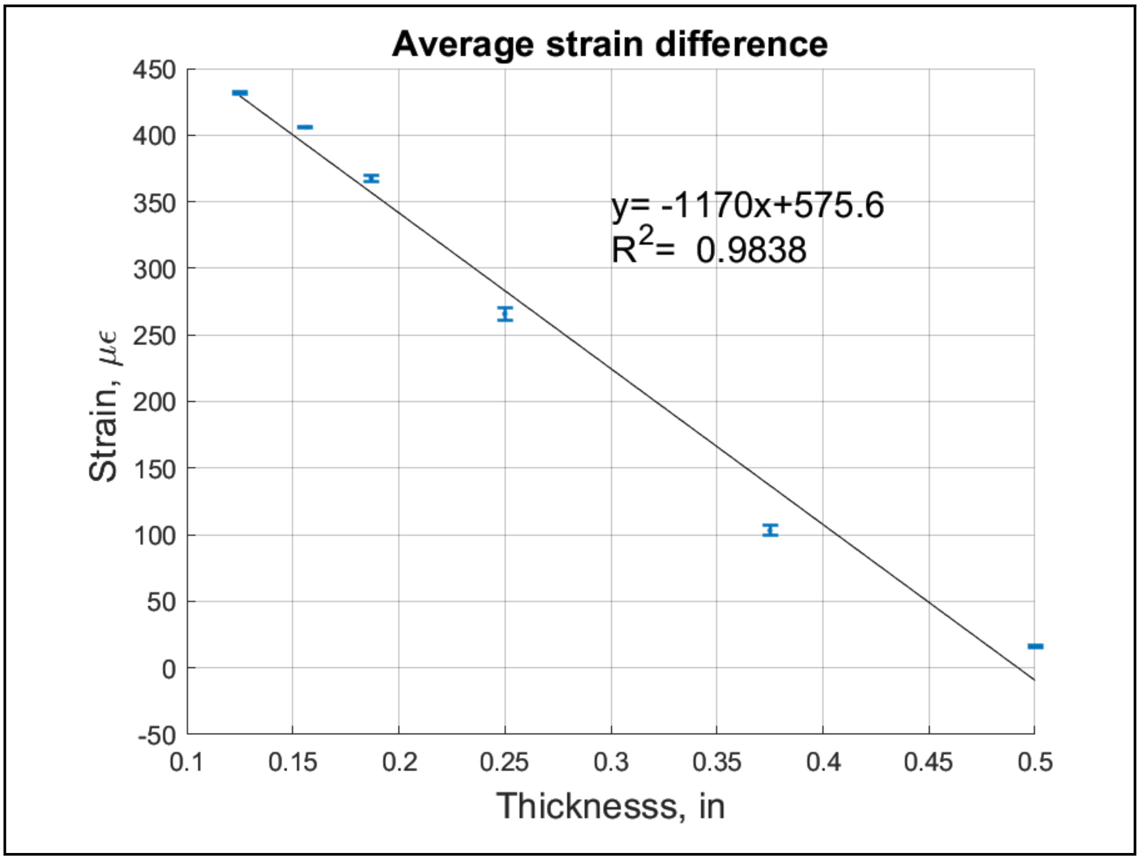

Figure 6: Graph of Average strain difference, Microstrain versus Thickness in inches

Appendix

Terfenol (Set Width) (Default Blue Line)

```
c_p= 330; %specific heat [J/(kg*K)]
rho= 9250; %density [kg/m^3]
L= 35*10^-3; %length [m]
W= 4*10^-3; %width [m]
calculations(L,W,rho,c_p)
```

Inconel (Variable Width) (Default Red Line)

```
c_p= 381; %specific heat [J/(kg*K)]
rho= 8610; %density [kg/m^3]
L= 35*10^-3; %length [m]
W= 3.85*10^-3; %width [m]
calculations(L,W,rho,c_p)
xlabel('Time, s')
ylabel('Temperature, K')
legend('Terfenol-D','Inconel Alloy 22')
```

```
function calculations(L,W,rho,c_p)
        T_a= 25; %ambient temperature in [°C]
        T_s= 20; %surface temperature in [°C]
        T_f=(T_a+T_s)/2; %film temperature [°C]
        T_f_K=T_f+273.15; %film temperature [°K]
        B=1/T_f_K; %beta [K^-1)

        k= 0.025325; %thermal conductivity [W/(m*K)]
        Pr= 0.73025; %Prandtl Number
        v= 1.539*10^-5; %kinematic viscosity [(m^2)/s]

        g=9.81; %acceleration due to gravity [m/s^2]

        V=L*W*W; %volume [m^3]
        D=sqrt((4*V)/(pi*L)); %equivalent diameter [m]
        Ra_D=((g*B*(T_a-T_s)*L^3)/v^2)*Pr; %Rayleigh Number
        Nu_n=0.387*Ra_D^(1/6); %Nussult Number Correlation Numerator
        Nu_d=(1+(0.559/Pr)^(9/16))^8/27; %Nussult Number Correlation Denominator
        Nu2=(0.6+Nu_n/Nu_d)^2; %Nussult Number Correlation
        h2=(Nu2*k)/D; %average heat transfer coefficient [W/(m*K)]
        A_s=4*(L*W)+2*(W*W); %surface area [m^2]
```

```
        Bi=(h2*V)/(k*A_s); %Biot Number "One may still wish to use lumped system
        %analysis even when the criterion Bi < 0.1 is not satisfied, if
        %high accuracy is not a major concern."
        b=(h2*A_s)/(rho*V*c_p); %[s^-1]

        t=0:0.1:2000; %time variable for plotting
        T=(exp(-b*t))*(T_s-T_a)+T_a+273.15; %Temperature as a function of time [°C]

        plot(t,T) %plotting the data
        hold on %to plot on the same figure
end
```

Figure A1: MATLAB code used to plot thermal response

<u>Uncertainty type B</u>

```
U1=1; % Picometer
U2=1; % Picometer
U3=1/1.2; %micro strain/Picometer;
UT= U3*sqrt(U1^2+U2^2) % micro strain
```

<u>0.500 Coupon uncertainty type A</u>

```
% U51 = 4.47134; %inconel micro strain
% U52 = 4.67818; %terfenol micro strain
% U5T = sqrt(U51^2+U52^2);
U5T = sqrt(1.2688^2);
```

Total uncertainty for 0.500 coupon

```
UT0500 = sqrt(U5T^2+UT^2)
```

0.500 coupon 95% confidence 16.12 +/- 3.46 microstrain

<u>0.375 Coupon uncertainty type A</u>

```
% U3751 = 4.34774; %inconel micro strain
% U3752 = 4.48724; %terfenol micro strain
% U3375T = sqrt(U3751^2+U3752^2);
U3375T = sqrt(3.6491^2);
```

Total uncertainty for 0.375 coupon

```
UT0375 = sqrt(U3375T^2+UT^2)
```

0.375 coupon 95% confidence 103.2 +/- 7.67 microstrain

0.25 Coupon uncertainty type A

```
% U251 = 6.67337; %inconel micro strain
% U252 = 5.72301; %terfenol micro strain
% U25T = sqrt(U251^2+U252^2);
U25T = sqrt( 4.8944^2);
```

Total uncertainty for 0.250 coupon

```
UT0250 = sqrt(U25T^2+UT^2)
```

0.25 coupon 95% confidence 265.9 +/- 10.07 microstrain

0.187 Coupon uncertainty type A

```
% U1871 = 5.49468; %inconel micro strain
% U1872 = 5.88628; %terfenol micro strain
% U187T = sqrt(U1871^2+U1872^2);
U187T = sqrt( 2.7215^2);
```

Total uncertainty for 0.187 coupon

```
UT0187 = sqrt(U187T^2+UT^2)
```

0.187 coupon 95% confidence 367.4 +/- 5.93 microstrain

0.156 Coupon uncertainty type A

```
% U1561 = 4.76604; %inconel micro strain;
% U1562 = 5.09837; %terfenol micro strain;
% U156T = sqrt(U1561^2+U1562^2);
U156T = sqrt(0.8245^2);
```

Total uncertainty for 0.156 coupon

```
UT0156 = sqrt(U156T^2+UT^2)
```

0.156 coupon 95% confidence 406 +/- 2.88 microstrain

0.125 Coupon uncertainty type A

```
% U1251 = 7.42481; %inconel micro strain;
% U1252 = 6.65039; %terfenol micro strain;
% U125T = sqrt(U1251^2+U1252^2);
U125T = sqrt(1.2134^2);
```

Total uncertainty for 0.125 coupon

```
UT0125 = sqrt(U125T^2+UT^2)
```

0.156 coupon 95% confidence 431.6 +/- 3.83 microstrain

Figure A2: MATLAB code used to calculate the Type B Uncertainty, Type A Uncertainty, and 95% Confidence

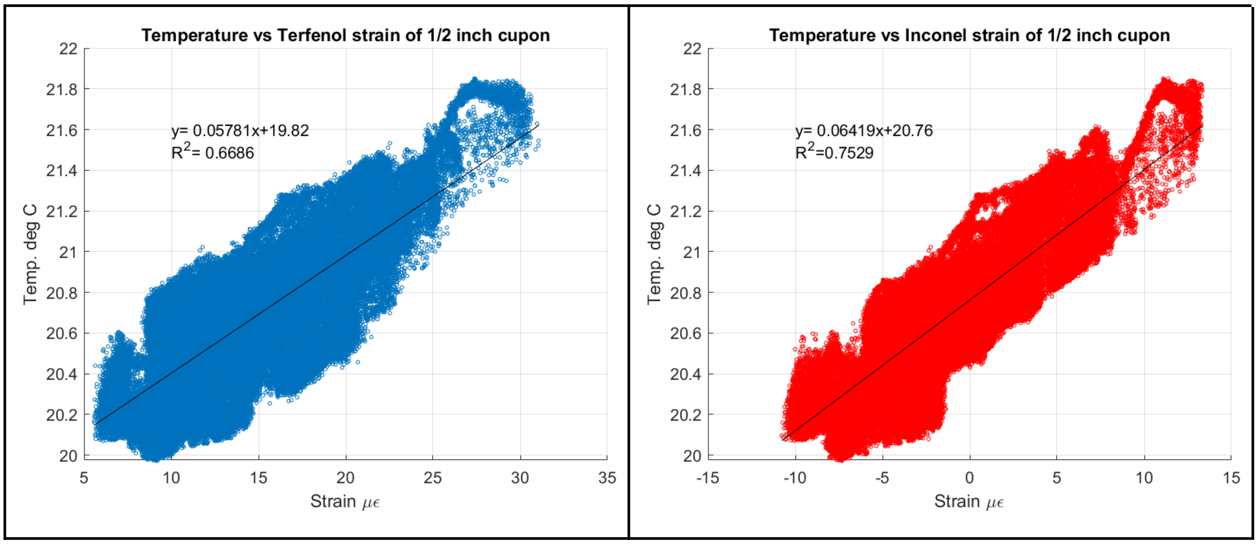

Figure A3: Temperature vs. Strain for 0.500 inch coupon

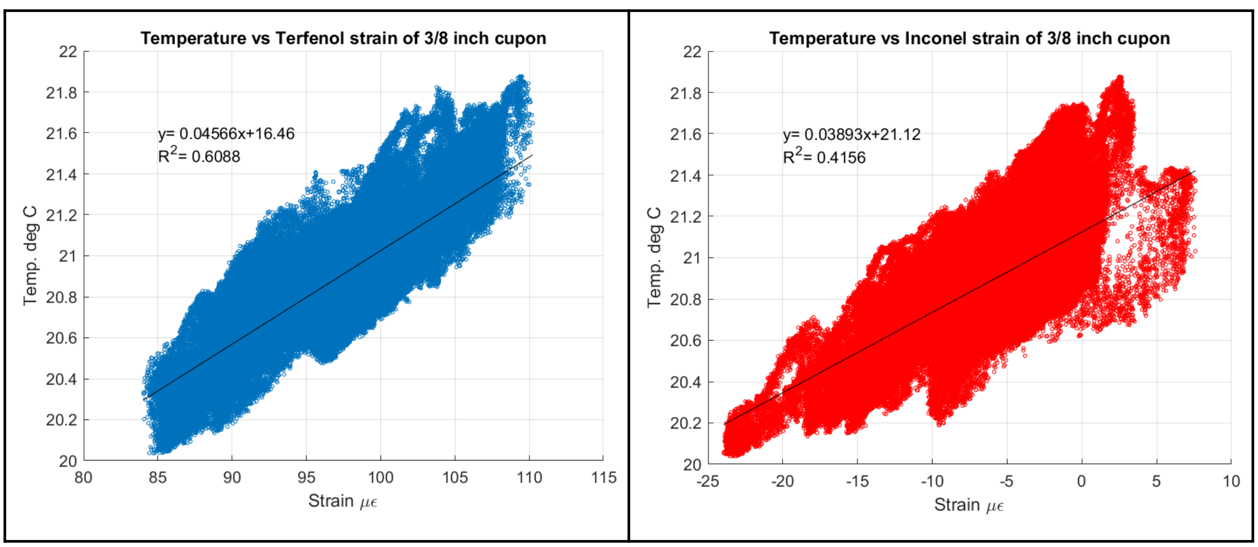

Figure A4: Temperature vs. Strain for 0.375 inch coupon

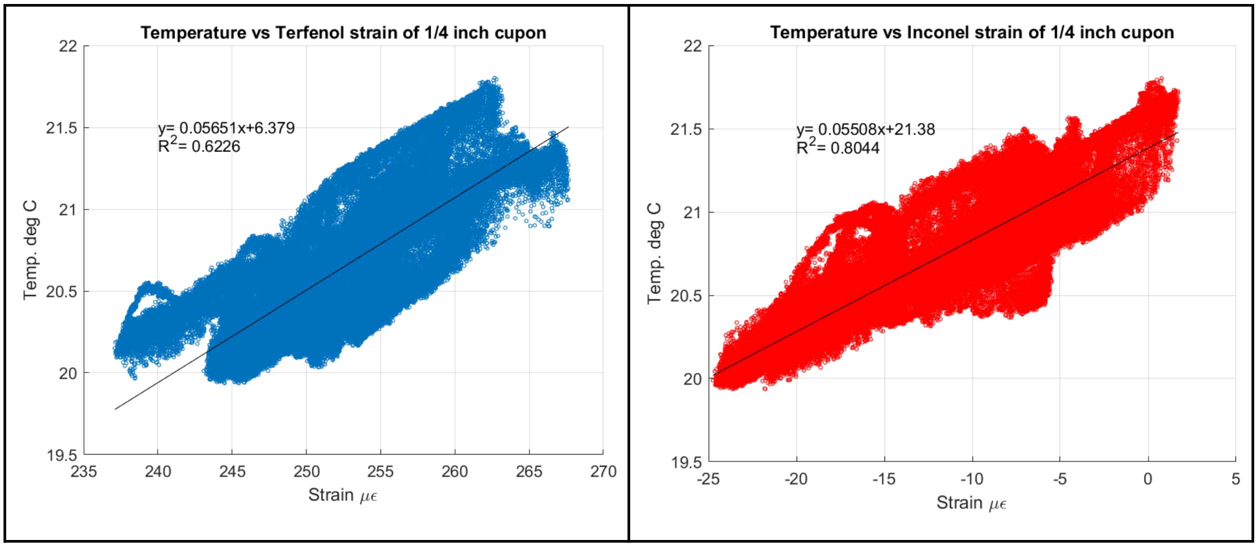

Figure A5: Temperature vs. Strain for 0.250 inch coupon

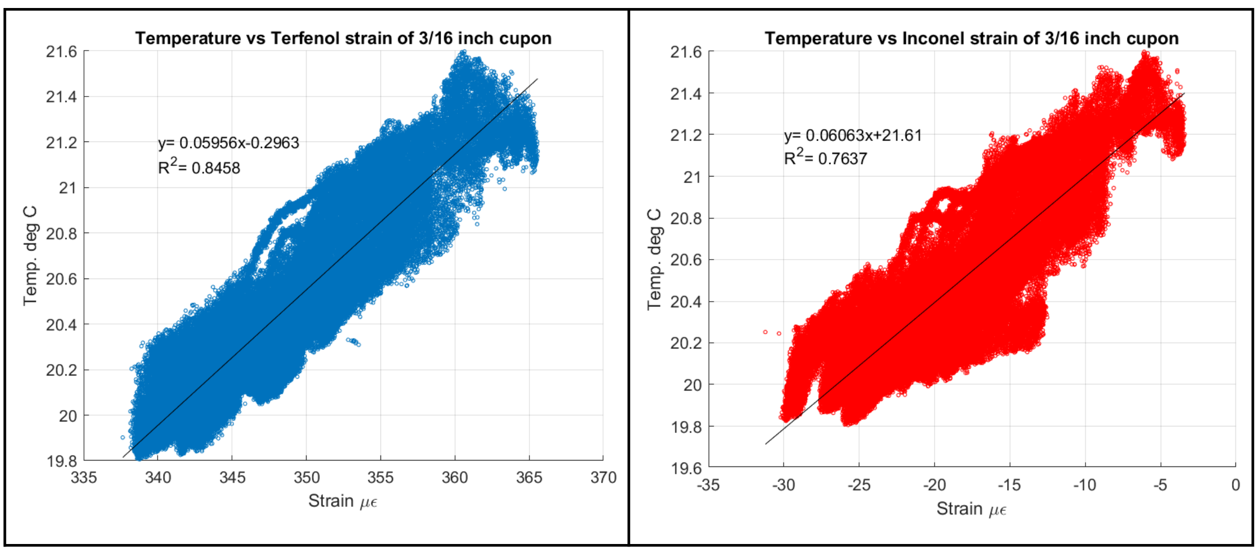

Figure A6: Temperature vs. Strain for 0.187 inch coupon

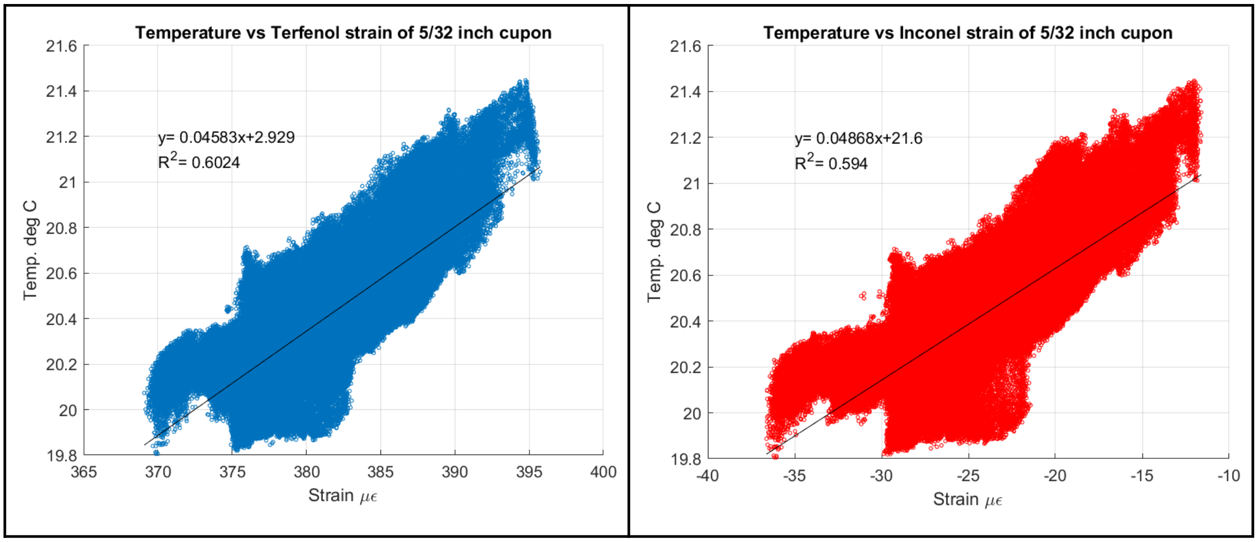

Figure A7: Temperature vs. Strain for 0.156 inch coupon

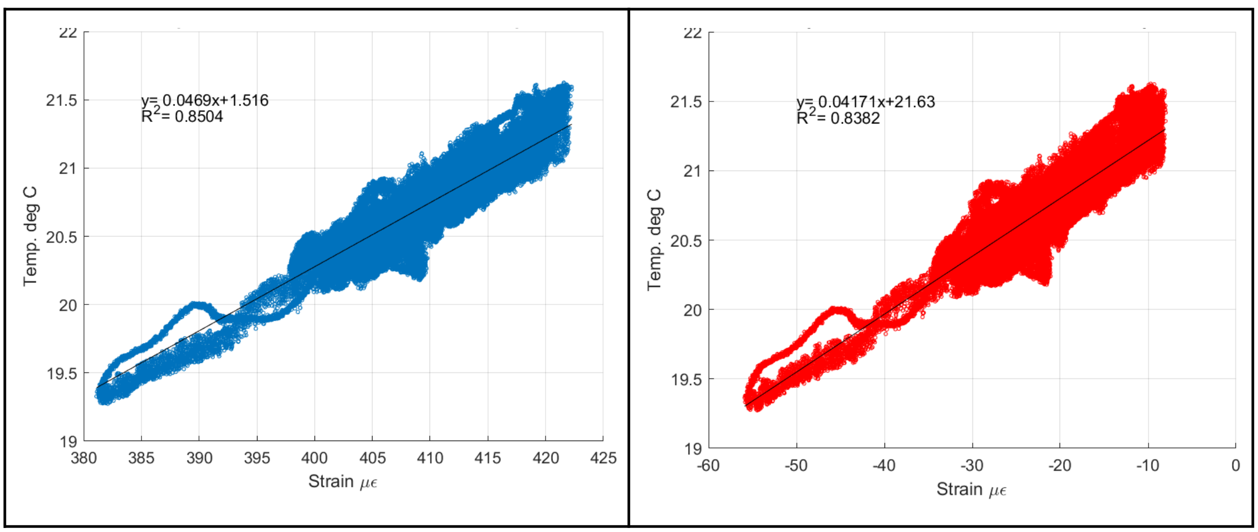

Figure A8: Temperature vs. Strain for 0.125 inch coupon

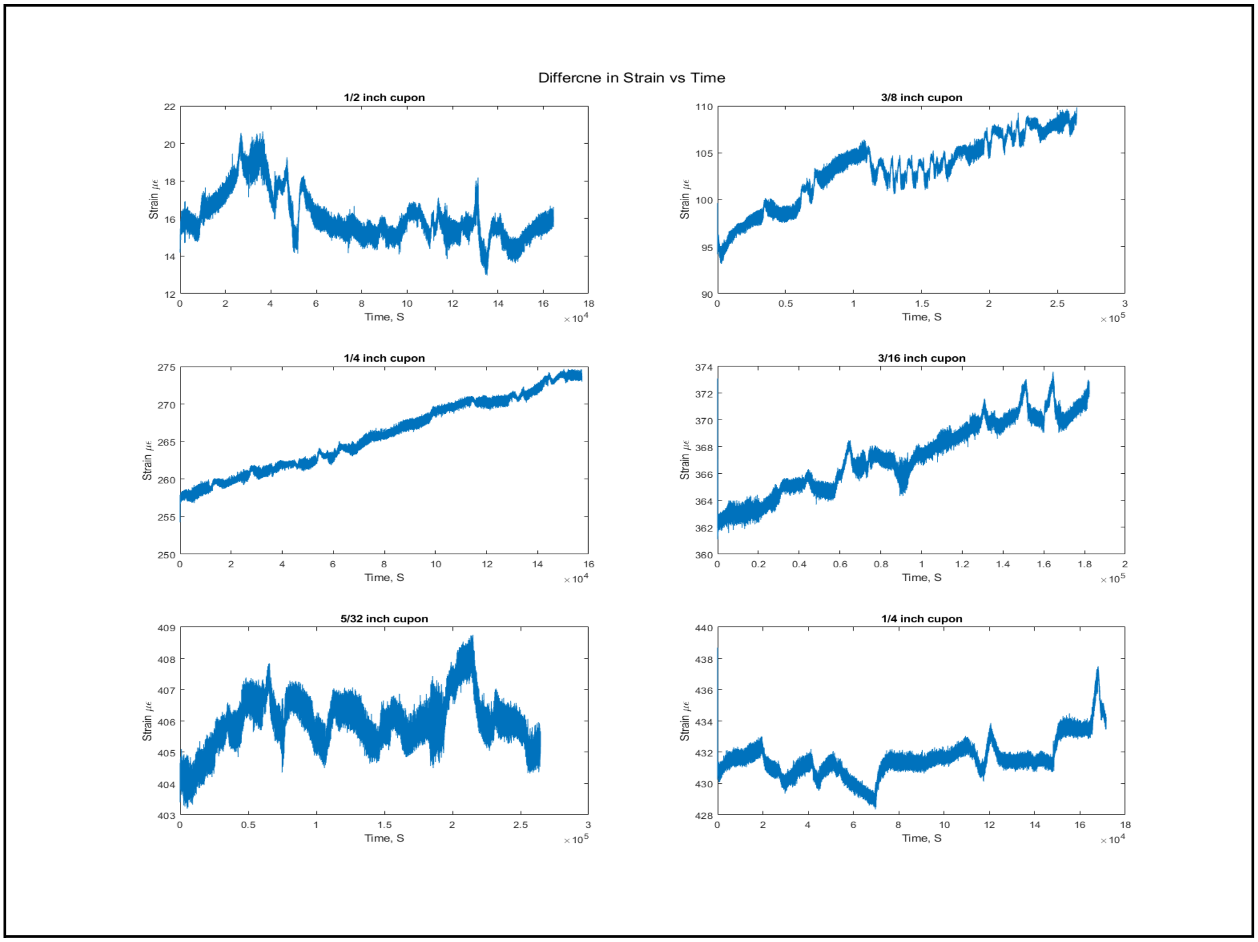

Figure 9: Plot of difference in strain of the two rods vs time for each coupon

References

[1] Luna, "si155 HYPERION Optical Sensing Instrument' si1255 REV.3, 09.21.21
[2] "Terfenol-D - Etrema Products, Inc.." *TdVib, LLC*, http://tdvib.com/terfenol-d/.
[3] *Alloy 22 Inconel - Special Metals Corporation*. https://www.specialmetals.com/documents/technical-bulletins/inconel/inconel-alloy-22.pdf.
[4] Çengel Yunus A., and Afshin J. Ghajar. *Heat and Mass Transfer: Fundamentals & Applications*. McGraw Hill Education, 2015.
[5] Luna, "os1100 Fiber Bragg Grating" FFP-SI REV.1, 12.27.19